pressure and significant aircraft response data were recorded on a wideband FM recording system used for real-time and spectral analyses.

The basic transonic maneuver was the wind-up turn executed at specific Mach number and starting altitude. The complete maneuver lasted from 12 to 18 seconds, which included level flight entry, buffet onset, wing rock, recovery, and the intermediate stages. The data acquired in a small time segment of the maneuver were assumed to be stationary so that auto- and cross-power spectra could be generated. The display of a set of power spectra generated for various time segments of the maneuver yielded results that illustrated the various phases of the buffet phenomenon.

To complement the flight test results, work was carried out to correlate the response data using the measured buffet pressure and analytical computation technique taking into account the aircraft structural and aerodynamic characteristics. The correlation procedure and other statistical processing techniques are described in the paper.

BUFFET FLIGHT TEST OF F-5A

The F-5A used in the buffet test was a single-seat fighter capable of carrying stores at wing fuselage pylon stations. The buffet test was conducted with two wingtip stores; otherwise the wing was clean. A three-view drawing of the F-5A is shown in Figure 1. The essential data of the wing are given below:

Airfoil Section	NACA 65A004.8 (Modified)
Area (Reference)	15.79 m^2 (170.00 ft^2)
Span (clean tips)	7.696 m (25.25 ft)
Aspect Ratio	3.75
Taper Ratio	.20
Sweepback (25% Chord)	24°
Mean Aerodynamic Chord	2.356 m (7.73 ft)
Dihedral Angle	0
Incidence Angle	0

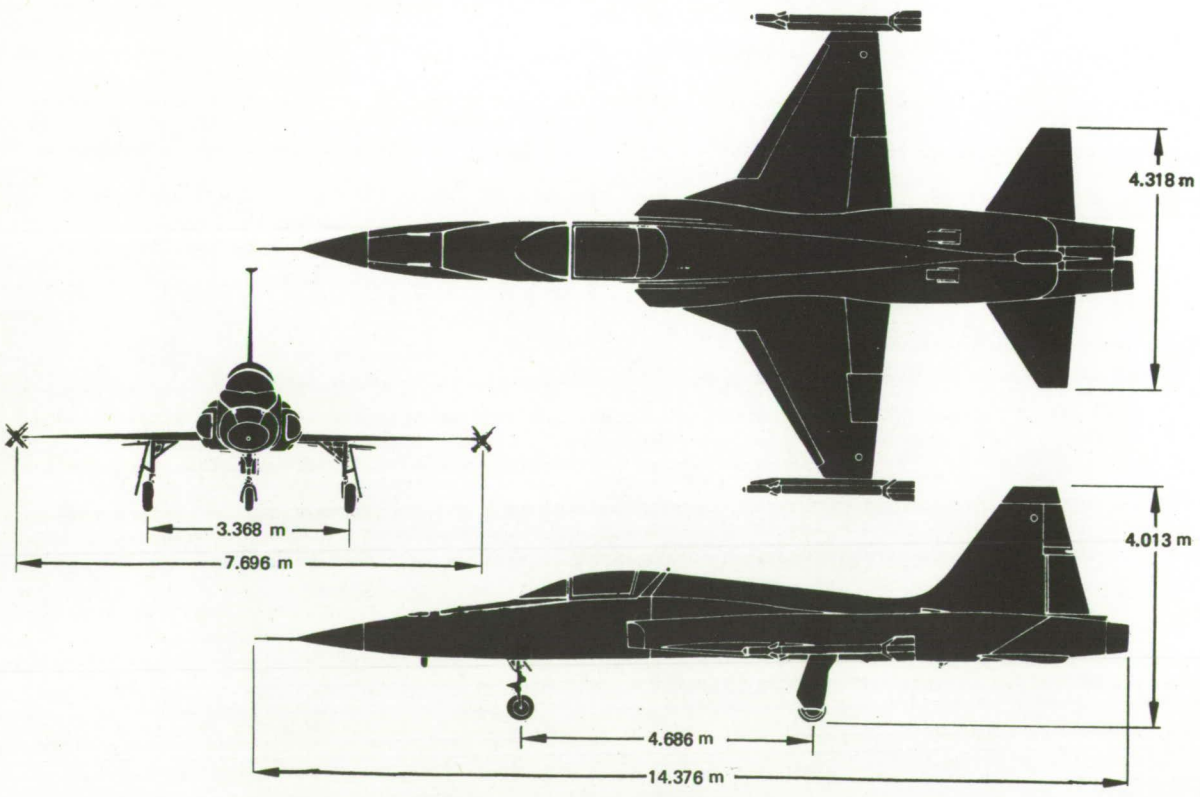


FIGURE 1. BASIC DIMENSIONS OF F-5A